

163
Phishing emails

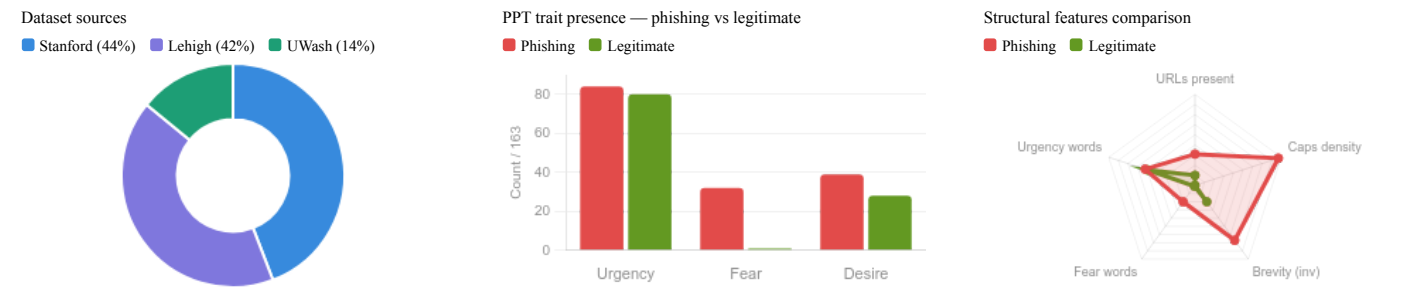
163
Legitimate emails

84
Avg word count (phish)

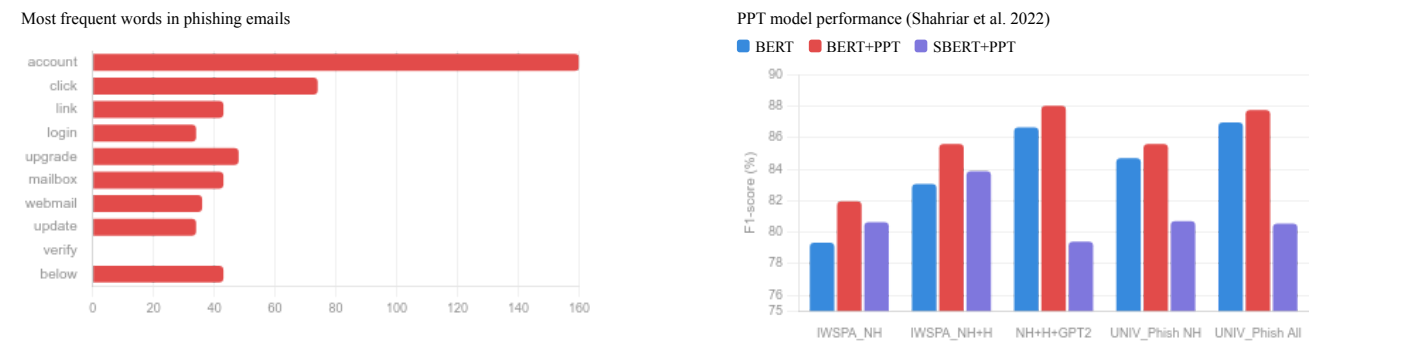
50
Phishing with live URLs

3
University sources

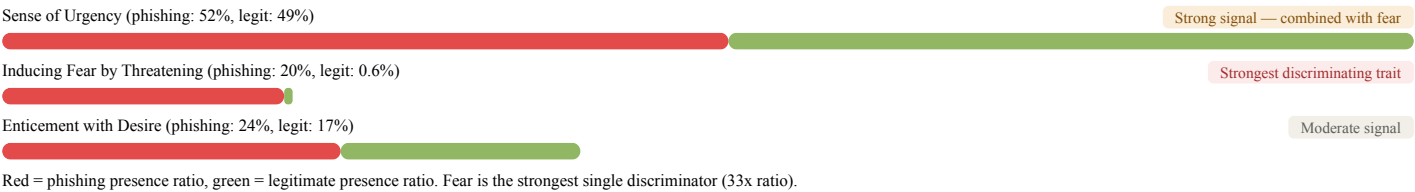
Dataset composition & psychological trait distribution



Top indicator word frequency (phishing emails)



Phishing Psychological Traits (PPT) signal strength



Sample phishing emails — annotated analysis

PayPal Account Verification Scam

From: security@paypal-support.com · Typosquat domain (paypal ≠ paypal)

"Your PayPal account will be locked. Click below immediately: http://secure-account-verify[.]com"

Fear: account lockout Urgency: "immediately" Malicious domain No SPF/DMARC (MXToolBox) No MX record found

Phishing confirmed

Microsoft Security Alert Spoof

From: support@microsoft-security.com · Character substitution (0 → o)

"We detected unusual login activity... verify your account immediately... Failure to verify within 12 hours may result in limited access."

Fear: account suspension Urgency: 12-hour deadline Brand impersonation Lookalike domain

Phishing confirmed

HDFC Bank Account Restriction Lure

From: alerts@secure-hdfcbank-support.net · Non-official .net TLD, embedded subdomain

"Your online banking access has been temporarily restricted. Confirm account details immediately: http://banking-verification-center[.]com"

Fear: banking restriction Urgency: "today" Credential harvesting URL Generic credential phishing

Phishing confirmed

"New Message from (801) 477-6790... Download attached file to listen to your voice message." → HTML attachment opens a fake Microsoft/Outlook login page harvesting credentials.

Curiosity lure (vmail) Credential harvest page HTML attachment vector Brand impersonation: O365 AiTM: pre-fills email

Indicators of compromise (IOC) summary

Domain typosquatting paypal-support.com, microsoft-security.com, hdfcbank-support.net — character substitution and brand name embedding in subdomain/path	No SPF/DMARC records MXToolBox confirms paypal-support.com has no SPF, no DMARC, no MX — domain created solely for phishing	HTML attachment payload audio_john_doe file is a local HTML page rendering a fake Microsoft login, pre-filling target email — AiTM pattern
Credential harvesting URLs secure-account-verify[.]com, microsoft-account-secure-check[.]com, banking-verification-center[.]com	High caps-word density Phishing emails average 2.17 ALL-CAPS words vs 0 in legitimate — a reliable corpus-level signal	Short email bodies Phishing avg 84 words vs 258 words legitimate (3× shorter) — minimal content maximises urgency-to-action ratio

Phishing taxonomy findings

CATEGORY	FINDING	SEVERITY
Domain	Typosquatted sender domains using character substitution (0→o, 1→l) and brand-name embedding in subdomain to bypass visual inspection	Critical
Auth	All sample phishing domains lack SPF, DMARC, and valid MX records — indicative of throwaway infrastructure with no email authentication	Critical
PPT: Fear	Fear trait has 33× higher rate in phishing (20%) vs legitimate (0.6%) — strongest single classifier signal per Shahriar et al. ablation study	High
PPT: Urgency	Urgency alone has low discriminating power (52% phish vs 49% legit) but when combined with Fear achieves 71.42% co-occurrence rate	Medium
Payload	HTML attachment phishing (voicemail lure) opens locally, bypasses URL scanners, pre-fills victim email for AiTM credential relay	High
ML Model	BERT+PPT achieves 88.04% F1 on IWSPA test set, outperforming SOTA by 4.54%. Fear PPT provides largest single-trait boost (per Fig. 4a)	Research finding
Dataset	UNIV_Phish (326 emails, balanced 50/50) from Stanford, Lehigh, UWash + ENRON ham — clean separation from IWSPA training set confirmed	Informational

Detection recommendations

Technical controls	User awareness signals
- Enforce SPF/DKIM/DMARC reject policy on inbound mail - Flag emails where sender domain has no MX record - Block/quarantine HTML attachments from external senders - Enable lookalike domain detection (typosquatting alerts) - Deploy BERT+PPT classifier on mail gateway (88% F1)	- Verify sender domain matches official brand domain exactly - Treat any email with Fear + Urgency + link as high-risk - Never open HTML attachments claiming to be audio/voicemail - Use MXToolBox to verify sender domain health - Check for missing brand logo (BIMI failure = no DMARC)

References

Shahriar, S., Mukherjee, A., & Gnawali, O. (2022). Improving Phishing Detection Via Psychological Trait Scoring. IADIS International Conference Web Based Communities 2022, pp. 131–139.

Dataset: UNIV_Phish — 163 phishing emails (Stanford, Lehigh, UWash) + 163 ENRON ham emails. Balanced, not overlapping with IWSPA training set.

MXToolBox Domain Health Check — paypal-support.com: 9 errors, 1 warning, 9 passed. No SPF, No DMARC, No MX, DNS not resolved.

HTML Attachment Sample — audio_john_doe_target_example_com_file.htm: renders fake Microsoft/Outlook login page, pre-fills john.doe@target.example.com.